

Development of Underwater Bot

B .Tech. Project

Submitted in partial fulfillment of the requirements

of the degree of

Bachelor of Technology

in

Electrical Engineering

By

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Declaration

We hereby declare that the project entitled “Development of Underwater bot” submitted in partial fulfillment for the award of the degree of Bachelor of Technology in ‘Electrical Engineering’ completed under the supervision of Prof. Srivathsan Vasudevan, Professor – Electrical Engineering Department, IIT Indore is an authentic work.

Further, we declare that we have not submitted this work for the award of any other degree elsewhere.

We declare that this written submission represents my ideas in our own words and where other’s ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misinterpreted or fabricated or falsified any idea/data/fact/source in my submission. We understand that any violation of the above will be cause for disciplinary action by the institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Giridhar A. P Tejas Chaudhari

Date:

Place: IIT Indore

Preface

This report on **“Development of Underwater Bot”** is prepared under the guidance of Prof. Srivathsan Vasudevan.

The project focuses on addressing the critical challenge of recovering submerged human bodies in low-visibility aquatic environments, a task that currently poses significant risks to manual divers. Through this report, we present the complete design and development of a cost-effective underwater robot capable of navigating these difficult conditions.

We have attempted to cover every aspect of the system’s prototyping, ranging from the mechanical design of the acrylic chassis and variable buoyancy mechanism to the embedded electronics and software pipeline. Special emphasis has been placed on the integration of the YOLOv8 machine learning model for real-time automated detection. We have documented our design choices, component selection, and experimental results to demonstrate that the design is both technically sound and feasible for future deployment in rural settings. We have also included detailed figures and 3D CAD models to illustrate the construction and assembly of the bot clearly.

Giridhar A. P, Tejas Chaudhari

B. Tech. IV year

Department of Electrical Engineering

B. Tech. Project Approval Certificate

This is to certify that the dissertation titled “**Development of Underwater bot**” submitted by **Giridhar A. P**, (Roll No. 220002033) & **Tejas Chaudhari**, (Roll No. 220002078) is approved for the award of degree of **Bachelor of Technology** in **Electrical Engineering**.

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Abstract

This project presents the design and development of a cost-effective underwater robot aimed at assisting in the search and recovery of submerged human bodies, particularly in rural Indian water bodies. Traditional recovery methods, such as manual diving, are often hindered by poor visibility and pose significant risks to human life. To address this, a tethered underwater bot was developed featuring a custom-built acrylic chassis and a 3D-printed internal structure. The control architecture utilizes a hierarchical setup, employing a Raspberry Pi 5 for high-level processing and an STM32 microcontroller for precise actuator management.

A key feature of the design is a syringe-based lead-screw mechanism for Variable Buoyancy (depth control), which operates alongside BLDC thrusters for maneuvering. For detection, the system integrates a low-light camera with a YOLOv8 machine learning model. To ensure real-time performance on edge hardware, the model was optimized to the NCNN format, achieving approximately 12 FPS. The prototype successfully demonstrates end-to-end functionality, including user-interface control, live video streaming, and real-time object detection. Future scope includes improving the waterproofing sealing for greater depths and implementing a UPS system for prolonged operation.

Keywords – Underwater Robotics, Computer Vision, YOLOv8, Dead Body Detection, Embedded Systems, CAD, Raspberry Pi, STM32.

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List of Abbreviations

Acronym	Definition
ABS	Acrylonitrile Butadiene Styrene
BLDC	Brushless DC
BMS	Battery Management System
CDC	Communication Device Class
CSI	Camera Serial Interface
ESC	Electronic Speed Controller
FPS	Frames Per Second
HAL	Hardware Abstraction Layer
LiPo	Lithium Polymer
PCB	Printed Circuit Board
PWM	Pulsed Width Modulation
ROV	Remote Operated Vehicle
RPi	Raspberry Pi
RTV	Room Temperature Vulcanizing
STM	ST Microelectronics
WHO	World Health Organization
YOLO	You Only Look Once

Chapter 1

Introduction

1.1 Introduction

Underwater environments are difficult to explore and monitor because of poor visibility, different depths, and the dangers of manual search operations. In complex situations, traditional techniques like diving or manual inspection are frequently unsafe and ineffective. The application of autonomous and remotely controlled underwater robots has grown in significance as a solution to these constraints. With increased precision and dependability, these systems can navigate underwater environments, record visual information, and identify objects of interest. The goal of this project is to create an underwater bot that can identify human bodies submerged in water. The bot offers a more secure and efficient alternative for manual search techniques by combining sensors, cameras, and control systems. The work represents a step toward applying robotics for real-world underwater detection and recovery operations.

1.2 Problem statement and Motivation for BTP

Problem Statement of the BTP is to develop an Underwater bot with underwater navigation and depth control for real time search operations that can accurately identify submerged human bodies in low visibility aquatic environments.

Although recovering submerged bodies underwater is extremely difficult, it is required in drowning, accident, and criminal cases. Traditional search techniques like manual diving have several major drawbacks, including poor visibility, depth restrictions, and a high risk to human life. According to the WHO, drowning claims the lives of nearly 300,000 people annually worldwide [1], and 30,000 people drown in India alone each year, with many of their bodies never

being recovered due to the nation's vast and complicated water bodies [2]. Recent advancements in robotics and AI-based underwater detection, such as enhanced YOLOv8 with image enhancement techniques, have shown encouraging accuracy (e.g., $\sim 94.7\%$) for small underwater objects [3]. These developments show how underwater bots can improve recovery efficacy and reduce human risk.

1.3 Organization of the Report

The remainder of this report is organized as follows. Chapter 2 presents a review of the existing literature regarding underwater object detection and the limitations of current autonomous solutions. Chapter 3 details the system hardware architecture, including component selection and the hierarchical control strategy. Chapter 4 outlines the software framework, covering the computer vision pipeline, user interface, and embedded firmware implementation. Chapter 5 discusses the mechanical design, specifically focusing on the chassis construction, waterproofing techniques, and the lead-screw mechanism. Finally, Chapter 6 summarizes the results obtained from the prototype and outlines the future scope of work.

Chapter 2

Literature Review

2.1 Introduction

Detection of submerged human bodies in low-visibility underwater environments is a difficult problem that combines robotics (autonomous navigation and depth control), sensing (optical cameras, sonar/acoustic sensors), image enhancement, and the machine learning. Severe light attenuation, color cast, turbidity, low contrast, and the lack of annotated datasets containing corpses or disfigured human bodies are the main challenges. Below is the summarized literature survey we conducted to tackle these challenges and develop a complete end to end underwater bot.

2.2 Summary

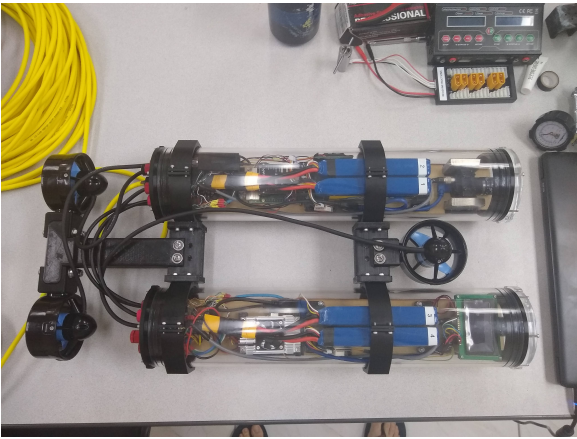
Due to poor visibility, light absorption, turbidity, and a lack of high-quality annotated datasets for identifying submerged human bodies, autonomous underwater search and recovery operations face significant challenges. A number of researchers use machine learning, acoustic sensing, and optical imaging to try to solve these problems. The successful realization of this project demands fulfillment of the strict constraints, particularly regarding cost and power.

The problem can be solved simply by collecting live video feed from a bot that dives underwater and running ML algorithms on a laptop connected to it [4]. This approach reduces the compute requirements of the bot, but it is very inconvenient for large-scale deployment where each laptop operating the bot is supposed to have the setup installed. Another approach makes use of acoustic solutions that can be fused with visual data for better results. Such systems are more complex, expensive, and power hungry; therefore, they are not feasible for the targeted

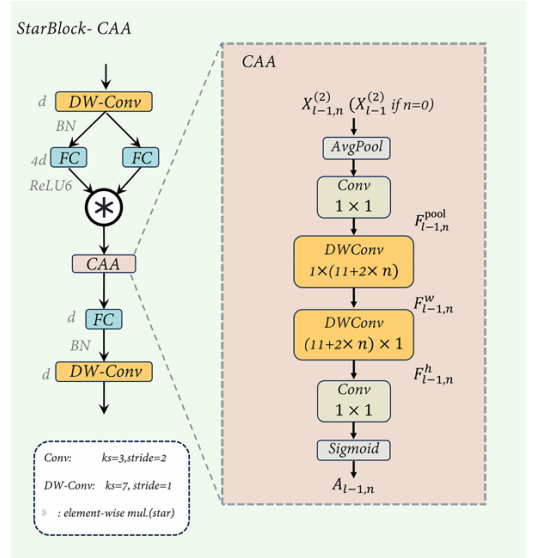
application.

The system architecture must make use of COTS, open-access, and modular components in order to get past the financial barrier. Marine Vehicle Packages (MVP), LoCO AUV [9], and MeCO (Medium Cost Open-source AUV) [7] are examples of frameworks that have emerged to offer adaptable platforms appropriate for research. The required flexibility is provided by these inexpensive designs, which frequently make use of parts like the BlueROV2 Base [8]. However, compared to pricey workclass vehicles, relying on these smaller, less expensive platforms comes with trade-offs, most notably limited depth ratings and a diminished capacity to withstand strong sea currents or waves. The Guidance, Navigation, and Control (GNC) subsystem must make up for this intrinsic stability limitation in the resulting system design.

Researchers have come up with a lightweight model that is capable of detecting human bodies from underwater images with very high detection accuracy rates [5]. This model has been trained to detect in various lighting conditions, at various distances, and even with multiple bodies in a single frame. The only drawback is that the model is trained only on online data consisting of living human beings, hence, the ability to detect disfigured human bodies is still uncertain. There are also various other models specially tuned to detect underwater objects from normal images such as Underwater-Yolo [6]. But it is not feasible to implement on a low-power device as in our use case.



(a) LoCO AUV



(b) Star Eye Model

Figure 2.1: (a) LoCO AUV and (b) Star Eye Model

Chapter 3

System Hardware Architecture

3.1 Hardware Components

A number of hardware components were utilized for motion control, depth control, visibility, and computation in order to make the prototype. The main parts used in the underwater bot are compiled in Table 3.1.

S. No.	Component	Specification	Purpose
1	2 × BLDC Motor + 2 × ESC	150A, 20A ESC, 3670 BLDC	Forward/Backward and Left/Right Motion
2	Stepper Motor + Driver	NEMA Stepper + A4988	Depth control via syringe lead screw mechanism
3	STM32F4 Blackpill	Cortex-M4 MCU	Motor and LED control
4	Raspberry Pi 5	Quad-core 2.4 GHz, 8 GB RAM	Server hosting, camera and ML integration
5	Limit Switches	Mechanical	Restrict syringe travel
6	LEDs	High-brightness LEDs	Illumination in turbid water
7	Li-ion/LiPo Battery	12 V DC	Powering all modules
8	Battery Management System (BMS)	40A, 12.6 V BMS	Overcharge and over-discharge protection
9	Sony IMX462 Arducam	Ultra Low-Light Camera	Underwater image/video capture

Table 3.1: List of hardware components used in the underwater human-detection robot

Each component has been selected based on performance requirements, and ease of integration with the rest of the system. The approximated cost required for making this project was around 45000 Rupees.

3.2 Control Unit

The underwater bot's hardware architecture is made to make sure that it can move around easily, control its depth accurately, and do fast real-time processing. A hierarchical control strategy is used to keep high-level image processing tasks separate from low-level control of the actuators.

- **High-Level Controller:** The **Raspberry Pi (RPi)** serves as the main computer. The Raspberry Pi uses the YOLOv8n model to detect humans in real time (as shown in the Fig. 3.1 (a)). The camera is a low light Arducam which connects via CSI interface to the RPi. The camera keeps sending live frames to the Pi, and the model analyzes each frame as it arrives. The RPi is very sensitive to its input voltage, hence we had to use a power bank for powering it up.

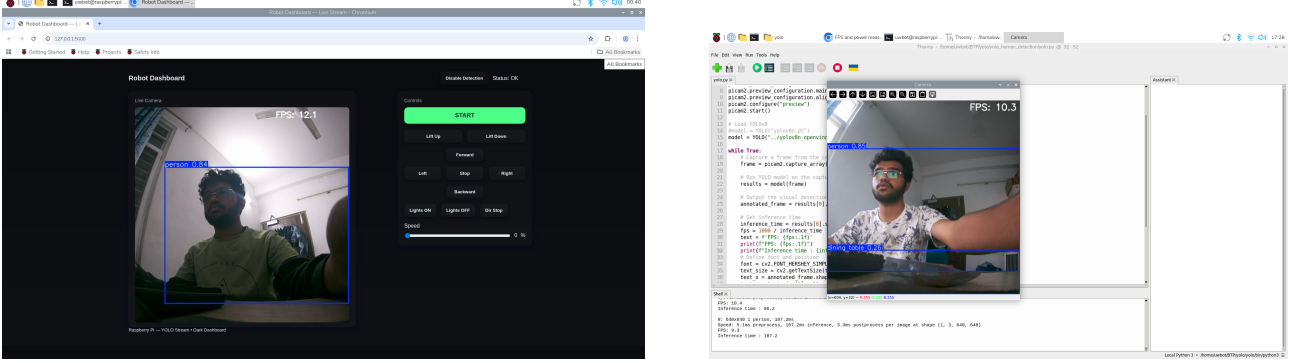
The RPi is also used for the hosting the web interface which has control functions for the bot, and live camera feed. Its local web interface served as the foundation for the bot's control system (as shown in Fig. 3.1 (b)). This interface provides the user with various control options, including forward, backward, up, down, left, right, start, stop, lights on/off and speed adjustment through a drag bar. In addition to that, it shows the live camera feed. The system also includes an option to enable or disable the human-detection feature. When enabled, each incoming frame is processed by the YOLOv8n. model for inference before being displayed. This inference can achieve around 10-20 FPS speed.

- **Low-Level Controller:** All electromechanical components are controlled in real time by an **STM32F4 Blackpill** board. It communicates with the stepper motor that powers the lead screw mechanism for depth control. The stepper is controlled by A4988 stepper motor driver. The limits of the lead screw are recognised by two limit switches. They trigger pin change interrupts when pressed preventing damage to the bot.

The main (one which controls the forward/backward motion) BLDC is a Rockstar RC boat motor with very high torque. It is driven by a FLYColour 150A Boat Electronic Speed Controller (ESC). The lateral BLDC (one which controls the right/left motion) is driven by a BLHeli based 20A ESC. BLHeli is a firmware for ESC. The timers of STM32 are configured to provide both the ESCs with appropriate PWM signals for establishing a proper control over the motion of the bot. The STM32 parses the incoming commands from the Raspberry Pi and converts them into instantaneous hardware actions. Robust communication with the RPi is ensured by using a custom Udev rule which operates over

the USB-CDC protocol.

There are two battery packs, one of 5000mAh (LiPo 21700) capacity for the stepper, LED and lateral BLDC motor. This is coupled with a 40A BMS for safe operation. The second one(1500mAh LiPo) provides for the main BLDC motor



(a) Web Interface

(b) YOLOv8n model Inference

Figure 3.1: (a) Web Interface and (b) YOLOv8n model Inference.

3.3 Communication Flow

- The laptop (ground station) connects to the RPi via a long Ethernet tether.
- The user accesses the web-based GUI from the laptop's browser to give commands and get the live camera feed to dead body detection.
- These commands are transmitted from the Raspberry Pi to the STM32 microcontroller over a USB (Serial) link using the USB-CDC protocol.
- The parses these commands and actuates the corresponding hardware.

Ethernet cable is pulled inside through resin for connecting to the RPi. We have used about 50m cable to ensure good accessibility of the bot. Ethernet was used because RF based solutions like WiFi and Bluetooth don't work under water.

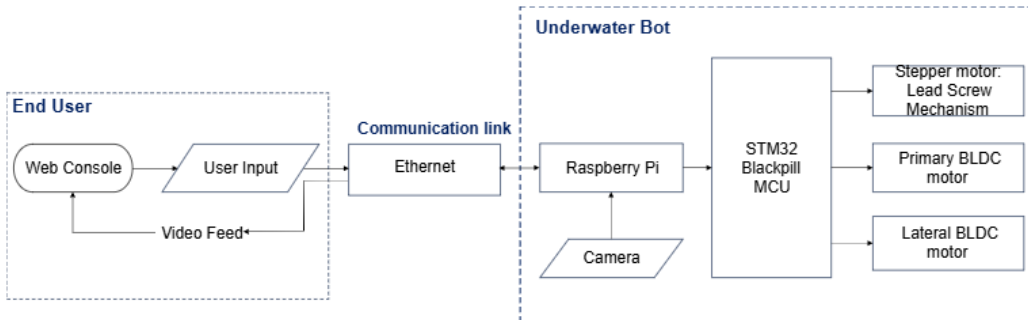


Figure 3.2: Communication and connection flow

Chapter 4

System Software Framework

The software stack is divided into four layers: the Computer Vision Pipeline, the High-Level User Interface(Host) , the Inter-Device communication bridge, and the Low-Level Embedded Control(Firmware). In the following sections, we discuss each of them in detail.

4.1 Computer Vision Pipeline

For achieving the dead body detection, after considering all available frameworks, we have settled with the YOLOv8n model. As the target is to run on RPi 5, we needed a very light weight model that could run with low power consumption and still be able to deliver enough frame rate.

The initial testing was done using YOLOv8n model and the frame rate achieved was approximately 6 FPS. To run it even more efficiently we converted it to the NCNN format. This boosted the frame rate to 12 FPS. It also helped to drop the power consumption which helped in increasing runtime of the overall system.

4.2 High-Level User Interface(Host)

The Raspberry Pi 5 locally hosts the web-based User Interface. To ensure smooth operation without need for manual setup, the system is configured for headless operation.

- **Auto-start configuration:** The web server is set to run on boot as a system service. This eliminates the need for external monitors or keyboards to start the device during field deployment.

- **Network Configuration:** The RPi is set to function as a DHCP server which assigns static IP address once connected via Ethernet. This ensures connectivity with any system having an Ethernet port.

4.3 Inter-Device Communication

The data that is fetched from the user via the UI needs to be reliably sent over to the STM32 for ensuring navigational safety. This is achieved in two steps:

- **Protocol:** Commands are transmitted via the USB-CDC (Communication Device Class) protocol. This acts like a virtual serial port, allowing for high speed low latency transmission of data packets while ensuring easy integration.
- **Device Identifier:** In Linux environments, USB Device identifiers can change upon reconnection. To prevent this, a custom Udev rule was written based on the STM32's unique vendor and product IDs. This guarantees the controller is always recognized by the same symbolic link (/dev/ttySTM32), preventing any runtime errors.

4.4 Low-Level Embedded Control(Firmware)

The motion control logic is executed on the STM32F411CEU6, developed within the STM32CubeIDE using the Hardware Abstraction Layer (HAL).

- **Bare-Metal Architecture:** The code runs on "bare metal" (without any Operating System like RTOS or Zephyr) to ensure deterministic behaviour. This guarantees that motor control loops are executed with precise timing, unaffected by OS background processes.
- **PWM Generation:** Two hardware timers of the STM32 are used to generate the necessary PWMs for both the ESCs as well as the LED lights. This offloads the CPU of the STM32 and ensures-jitter free signal for the ESCs.
- **BLHeli Configuration:** The lateral thruster's ESC (8-bit) was flashed and tuned using BLHeliSuite software. This was used to enable bidirectional motion of the BLDC and to calibrate it for the output PWM range of the STM32. This allows the BLDC motors to reverse instantly and enables quick braking . It also removes dead zones in the throttle response of the ESC.

Chapter 5

Mechanical Design

This chapter explains how the bot's physical structure was designed and built. It covers the main body construction, the custom 3D-printed parts, the waterproofing methods, and the internal mechanical layout. The aim is to show how the bot's body was made, watertight, and capable of storing all components effectively, while also supporting lead screw mechanism.

5.1 Chassis Construction

The bot's main body is built using a transparent acrylic tube that measures 60 cm in length and 5 inches in diameter. The tube provides a rigid, lightweight, and visually accessible enclosure, making it easy to observe internal components during assembly and testing. Acrylic was chosen because of its strength, transparency, and excellent waterproofing characteristics when properly sealed.

Because commercially available end caps for this tube size were not available, custom 3D-printed end caps were designed. These end caps were dimensioned precisely to match the tube's dimensions, ensuring a reliable and tight fit.

5.2 Waterproofing and 3D-Printed Components

The custom end caps were designed with O-ring slots, following the standard O-ring design guidelines [10], to ensure effective water sealing. The 3D-printed parts were printed using ABS material (as shown in Fig. 5.1) with 20 infill, providing a good balance between durability and flexibility. These ABS printed were smoothen using Acetone, so that water doesn't leaks

through the small pores. All the 3d printed parts were designed using Autodesk Fusion 360 (Fig. 5.2). To improve water resistance, the printed parts were coated with a thin layer of clear resin, which also helps reduce small pores typically present in 3d prints. Water inlet ports for the syringe mechanism were implemented using pneumatic couplers. For additional sealing, Teflon tape and RTV silicone were applied wherever necessary, especially around threaded or pressure areas. The front end of the bot contains a small window for the camera. This opening was sealed using a circular acrylic sheet filled with clear resin, ensuring clarity for the camera feed while maintaining waterproof integrity.



(a) 3d printed parts



(b) End cap assembled

Figure 5.1: (a) 3d printed internal parts and (b) End cap assembled.

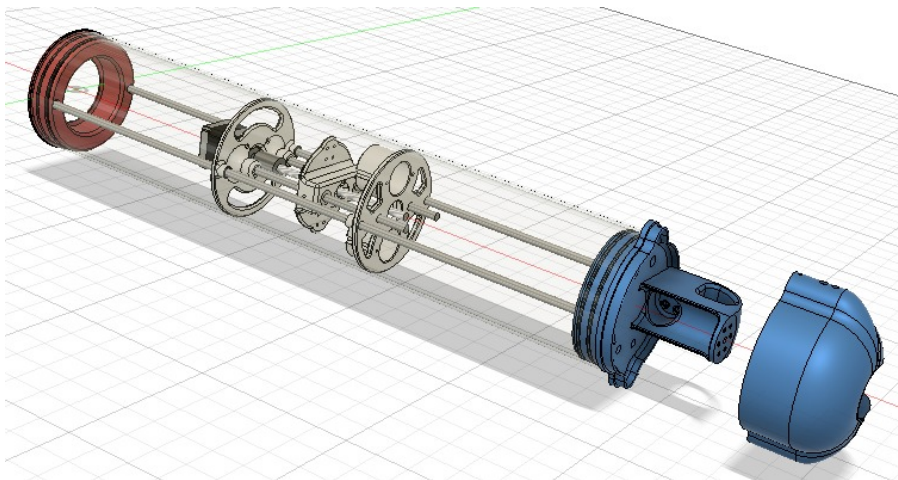


Figure 5.2: Fusion model of the Bot

5.3 Internal Mechanical Layout and Lead-Screw Mechanism

To ensure that internal components remain fixed in place, the bot uses two steel guiding rods with circular mounting stages. These rods have dual purpose: they make the internal assembly of the bot stable and also contribute additional weight, helping the bot to achieve neutral buoyancy.

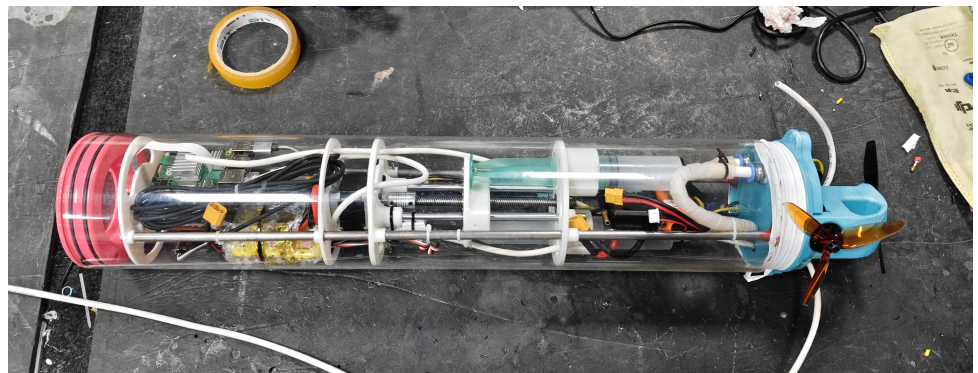
The central region of the bot contains the syringe based actuation system for the height control, which is driven by a stepper motor through a lead screw mechanism (as shown in Fig. 5.3 (a)). The lead screw plays a crucial role in converting the rotational motion of the stepper motor into smooth linear displacement of the syringe plungers. This mechanism offers several advantages:

- High precision in controlling linear motion.
- Inherent mechanical advantage, reducing load on the motor.
- Smooth movement, useful for consistent fluid control.
- Self locking tendency, preventing back driving when the motor is not powered.

This setup ensures reliable operation even in the underwater pressure variations.



(a) Lead Screw Mechanism



(b) Final Assembled Bot

Figure 5.3: (a) Lead Screw Mechanism and (b) Final Assembled Bot.

Chapter 6

Results, Conclusion and Future Works

6.1 Results & Conclusion

The current prototype is electronically fully functional. It is able to take user input from the user via web interface, convert it into control signals for the various actuators all while running dead body detection in the background. The ML model is able to send live camera feed with minimal latency. Demo video of the prototype and updated codes are available in **GitHub**.

6.2 Future scope of work

- Water leakage is there to a small extent which needs to be fixed. Afterwards end to end deployment can be done in water.
- A heatsink mechanism for cooling the RPi would help to extend the up-time of the bot. A wireless charging setup will enable us to permanently close the bot.
- Once the end to end testing is performed, a floating wifi router unit can be made. This will extend the range of the bot and make it more accessible for search operations.
- We can convert the user interface into a responsive web page. This will enable users with any screen size to control the bot after connecting to the WiFi router. This will allow for direct deployment in a rural setting.

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